You must not read the question paper before the exam starts.

You must not write anything until the invigilator says that you may start.

Pay attention to the instructions given by the invigilators.

Read the instructions below **carefully** before the exam starts, **check** you know how you are required to complete your answer booklets or sheets.

You must ensure that the front of your answer books/sheets have been completed correctly **by the end of your exam finish time,** no further time will be given.

January 2026

EC148 2025/6 A 800

Answer booklet
Calculators not permitted

University of Reading

Introductory Economics (EC148)

Three hours

Answer **all three** questions.

For full marks you need to justify your answers.

This paper carries a total of 100 marks.

Section A

1. Microeconomics

   Essay topic (1000 words): How have mathematical methods contributed to the development of microeconomic theory?

   Give one or more examples to illustrate your answer.

   (50 marks)

Section B

2. Macroeconomics

   Keynesian Short-Run Model

   (a) Briefly explain the following concepts: use words, graphs, and equations as necessary.

      (i) Balanced Budget

      (ii) Keynesian Cross Diagram

      Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

      (iii) When income is low and more is being spent on consumption than is being received in income, then the MPC must be greater than 1.

   (5 marks)

Question continued over leaf

(b) Consider the following macroeconomic model (where r is the percentage real rate of interest.)

Goods market:

$C = 1000 + 0.8Y_D - 1{,}000P$

$I = \frac{20{,}000}{3} - \frac{1{,}000}{3}r$

$T = 625$

$G = 2{,}000$

$NX = 500$

Money market:

$M^D = 20{,}000 - 2{,}000r$

$M^S = 10{,}000$

$P = 1$

Where *C* is consumption, Y_D disposable income, *P* the price level, M^D money demand, *I* investment, *r* the nominal interest rate, M^S money supply, *T* tax revenues, *G* government expenditure and *NX* net exports.

(i) Derive the equation for the LM curve. Illustrate equilibrium in the money market.

(ii) Derive the equation for the IS curve. Illustrate equilibrium in the goods market.

(iii) Solve for all equilibrium variables in this economy. Illustrate equilibrium using an IS-LM graph. What differentiates this graph from a standard IS-LM graph? Briefly explain.

(iv) Suppose the government wanted to balance its budget by adjusting taxes. What would it have to do (be precise) to achieve this? Briefly explain in words and using graphs the impact of this adjustment on your answer from part (iii). No equations are required to answer this question.

(20 marks)

3. Keynesian Long-Run Model

(a) Briefly explain the following concepts: use words, graphs, and equations as necessary.

(i) Full employment level of output

(ii) Cost-push Inflation

Indicate whether the following statement is true, false, or uncertain. Briefly explain your answer.

(iii) If aggregate demand decreases and neither short-run nor long-run aggregate supply changes, then the price level increases in the short-run and decreases in the long-run.

(5 marks)

(b) Consider an economy which is characterized by the following information. The full-employment level of output is $3{,}000$. The short-run supply of goods is given by $Y = 1500P - 6{,}000$ and aggregate demand is given by $Y = 18{,}000 - 1500P$.

(i) Given the above information, find the short-run equilibrium level of real GDP and the aggregate price level. Illustrate equilibrium (Be Precise and Label Everything). Is this economy in a recession or boom in the short run? Briefly explain your answer.

(ii) Calculate the long-run equilibrium level of real GDP and the aggregate price level. Briefly explain in words how we get from the short-run equilibrium to the long-run equilibrium. Illustrate this graphically.

(iii) After the economy returns to long-run equilibrium, what are the equations for the LRAS curve, the SRAS curve, and the AD curve? Briefly explain how you obtain these equations.

(20 marks)

(End of Question Paper)